Interview Prep: BCI Technologies — Control Systems Hardware Designer

Interview Date: February 19, 2026 **Interviewer:** Gil Isaiah, Vice President at BCI Technologies **Format:** Likely second/follow-up interview (first call was Jan 27) **Location:** TBD (Fort Myers, FL area or remote)

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1. Company Intel

BCI Technologies (Blackburn Controls Inc.)

- **Founded:** 1990 — over 35 years in business
- **HQ:** 1619 E. Vine Street, Kissimmee, FL 34744
- **Parent:** U.S. Water Services Corporation (New Port Richey, FL, 700+ employees, 20 states)
- **Revenue:** ~$7.4M (BCI entity)
- **Specialty:** Advanced automation and control solutions for **water & wastewater utilities**
- **What they build:** SCADA systems, telemetry networks, control panels, lift station controllers

Key Certifications & Partnerships

- **UL 508A certified panel shop** — designs, fabricates, and tests industrial control panels
- **Exclusive GE Channel Partner** for water/wastewater telemetry in Southeastern US
- **3,000+ GE MDS radios deployed**
- Certified on: **VTScada, EcoStruxure Geo SCADA, Aveva CitectSCADA, Ignition**

Products & Services

Category	Details
Systems Integration	Custom panel design, PLC & SCADA programming, end-to-end integration
Wireless Telemetry	GE MDS radio solutions for remote utility sites
LsPRO	Packaged lift station controller — their commercial product
SCADAaaS	SCADA-as-a-Service (cloud-hosted monitoring)

Recent Project (Mention This!)

Cape Coral Water Utility — $201K contract awarded to BCI (Dec 2024)

- Scope: Telemetry communication + SCADA integration for Cape Coral's wastewater and irrigation systems
- Scale: 24 master pump stations, 274 lift stations, 6 canal pump stations, reclaimed water systems
- Multi-phase replacement of obsolete systems
- **This is in YOUR backyard** (Naples/Fort Myers area) — great talking point

Leadership

Name	Role	Background
John Gilbert Blackburn	President, Treasurer, Director	Founder
Gil Blackburn	Semi-Retired SVP	Founder family
Gil Isaiah	VP (your interviewer)	30+ years SCADA experience, large system architecture, diesel generator controls. BS Mechanical Engineering from UF. Certified CitectSCADA Engineer.

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2. The Role

Control Systems Hardware Designer

Core Responsibilities:

1. **Read and understand project plans and specifications** — both control system and instrumentation
2. **Create control panel electrical schematics & panel layouts** using AutoCAD Electrical
3. **Assemble submittal packages** using PDF indexed format
4. Design and create submittal packages for projects

Translation: You're the person who takes an engineer's specs/drawings for a water utility project and turns them into production-ready panel schematics, wiring diagrams, and organized documentation that gets submitted to the client for approval before fabrication begins.

What a Submittal Package Contains

- Control panel electrical schematics (one-line diagrams, wiring diagrams)
- Panel layout drawings (physical arrangement of components)
- Bill of materials
- Data sheets for major components (PLCs, drives, instruments)
- Sequence of operations
- All organized in indexed PDF format

Tech Stack to Know

Tool	Your Experience
AutoCAD Electrical	Honest: SolidWorks primary, AutoCAD basics. Emphasize transferable skills + willingness to learn fast
PLC platforms	Allen-Bradley, Schneider likely. You have PLC + ladder logic from FGCU TA role
SCADA platforms	VTScada, Geo SCADA, CitectSCADA, Ignition — mention awareness
UL 508A standards	Know what this is: standard for industrial control panels (fire/shock safety). BCI is certified
EtherNet/IP	Direct experience from AXI International

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3. Your Interviewer: Gil Isaiah

- **Title:** Vice President, BCI Technologies
- **Experience:** 30+ years in SCADA system architecture and diesel engine generator control systems
- **Education:** BS Mechanical Engineering, University of Florida
- **Certification:** CitectSCADA Engineer
- **Responsibilities:** Business development, project management, software development
- **Style:** Likely technical + business-oriented. He's a VP who still does hands-on SCADA architecture — expect both technical depth and "will this person fit the team" questions

How to Connect with Gil

- He's a UF Mech-E — you're a FGCU Biomedical-E. Both Florida engineers. Both hands-on.
- He's been doing this 30+ years — show respect for the craft, ask about his career path
- He handles business development too — he'll appreciate someone who understands the business side

- **This is your SECOND interaction** — he already liked you enough from the first call to bring you back

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4. Your Strongest Selling Points

1. Controls Engineering TA (3 years)

This is your **#1 asset** for this role. You:

- Designed electrical schematics and wiring diagrams
- Built actual control panels with PLCs, sensors, instrumentation
- Created documentation packages (schematics, wiring diagrams, test procedures)
- Taught PLC programming, ladder logic, control systems to 100+ students
- If you can teach it, you understand it deeply

2. AXI International (Control System Layouts)

- Designed control system layouts for automated filtration in SolidWorks
- Created electrical schematics for sensor feedback loops
- PLC troubleshooting + EtherNet/IP exposure
- Industrial automation context — directly relevant

3. Regulated Documentation Experience

- Insulet (Omnipod): FDA-regulated submittal documentation, indexed packages
- Sodexo (Hospital): ISO 13485 compliance, reading electrical schematics for medical devices
- Marmon A&D: Military-grade cable documentation, VNA testing, project specs

4. Local to SW Florida

- Based in Naples — 2 hours from Kissimmee HQ, right next to Fort Myers project sites
- Familiar with Florida utility landscape
- The Cape Coral project is literally in your area

5. Engineering Mindset + Fast Learner

- BS Biomedical Engineering, 3.8 GPA, Honors
- Self-taught Python, automation, AI — shows ability to pick up new tools quickly
- AutoCAD Electrical is learnable in weeks for someone with your SolidWorks + electrical background

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5. Addressing the AutoCAD Electrical Gap

This is the elephant in the room. Here's how to handle it:

The Frame: "My CAD experience is primarily in SolidWorks, but the underlying skills transfer directly — I've created electrical schematics, control panel layouts, and wiring diagrams. AutoCAD Electrical is a tool I can learn quickly because I already understand what the output needs to look like. I've taught myself Python, AI tools, and multiple CAD platforms before — my track record shows I pick up new software fast."

Bonus: If you have time before the interview, download AutoCAD Electrical trial and spend 2-3 hours watching YouTube tutorials on:

- Creating electrical schematics in AutoCAD Electrical
- Panel layout drawings
- Wire numbering and component tagging
- PDF export/publishing

Even basic familiarity will show initiative.

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6. Likely Interview Questions & Answers

Technical Questions

Q: Walk me through your experience with control panel design.

> *"As a Controls Engineering TA at FGCU for three years, I designed and built control panels from scratch for our lab exercises. I'd start with the electrical schematics — power distribution, PLC wiring, sensor integration — then physically assemble the panels and create the documentation package. I built panels that included PLCs, temperature sensors, traffic light controllers, and PID control systems. Each one required a complete documentation package with wiring diagrams, component specs, and test procedures — very similar to the submittal packages you produce here at BCI."*

Q: What's your experience with AutoCAD Electrical?

> *"My primary CAD experience is in SolidWorks, where I've done both 3D modeling and 2D technical drawings including control system layouts and electrical schematics. I've used AutoCAD for basic work but haven't done extensive work specifically in AutoCAD Electrical. That said, I understand electrical schematic standards, component symbols, wire numbering conventions, and what a complete control system drawing set needs to contain. The transition to AutoCAD Electrical is really about learning the software interface — the engineering knowledge is already there. I'm committed to getting up to speed quickly, and honestly, for someone who's taught themselves Python, AI systems, and multiple engineering*

> *tools, learning a new CAD platform is straightforward."*

Q: What do you know about UL 508A?

> *"UL 508A is the standard for industrial control panels — it covers the safety requirements for panel construction, including things like proper wire sizing, short circuit current ratings, enclosure requirements, spacing, and overcurrent protection. I know BCI is a UL 508A certified panel shop, which means every panel you build has to meet those standards. My experience building control panels in the university lab gave me a foundation in these principles, though I'm sure there's more detail I'll learn about the specific UL requirements in a production environment."*

Q: Tell me about your experience reading project specifications.

> *"I've read and interpreted specs in multiple contexts. At AXI International, I worked from project specifications for industrial automation systems to design control system layouts. At Marmon, I interpreted electrical schematics and specs for military-grade cable production. At Insulet, I worked within FDA-regulated specifications for manufacturing equipment. And as a TA, I created specs for lab exercises and also taught students how to read and interpret technical drawings. I'm comfortable taking a spec set and translating it into actionable design work."*

Q: What PLC platforms have you worked with?

> *"I programmed PLCs using Arduino-based platforms for our controls lab — building state machines, traffic light controllers, temperature control systems with PID. I understand ladder logic, sequential function charts, and the fundamentals of PLC programming. At AXI, I supported troubleshooting on their PLC systems and worked with EtherNet/IP for data acquisition. I haven't worked extensively with specific industrial PLCs like Allen-Bradley or Schneider, but the programming concepts transfer directly. I'd need to learn the specific platform you use, but the logic and structure are the same."*

Q: What SCADA platforms are you familiar with?

> *"I'm aware of the major platforms — VTScada, Ignition, CitectSCADA, Wonderware. I know BCI is certified across several of these. My direct hands-on SCADA experience is limited, but I understand the architecture: field instruments and PLCs at the bottom, communication networks in the middle, and the SCADA HMI at the top for monitoring and control. For this hardware designer role, my understanding is the focus is more on the panel design and documentation side rather than SCADA programming, but I'm eager to learn the full picture."*

Behavioral / Fit Questions

Q: Why are you interested in this role / BCI specifically?

> *"A few reasons. First, I genuinely enjoy control systems work — it was my favorite subject at FGCU, and I spent three years teaching it. Second, water and wastewater infrastructure is critical work that matters. I saw you just won the Cape Coral telemetry project — 274 lift stations, multi-phase SCADA replacement — that's exactly the kind of work I want to be part of. Third, I'm based in Naples, so I'm local to your Florida project sites. And honestly, after speaking with you last time, the company culture and the hands-on technical work really appealed to me."*

Q: Where do you see yourself in 3-5 years?

> *"Growing within the controls and automation space. In this role, I'd want to become an expert at panel design and submittal documentation, then gradually expand into SCADA programming, project management, or systems architecture. I'm the kind of person who wants to understand the whole system, not just my piece of it. Long-term, I'd love to be someone like the engineers here who can take a project from concept to commissioning."*

Q: Tell me about yourself.

> *"I'm a biomedical engineer from FGCU — graduated honors with a 3.8 GPA. What most people don't expect is that my favorite part of the curriculum was controls engineering. I was a TA for three years — designed and built control panels, taught PLC programming to 100+ students, and created all the lab documentation. After graduating, I gained industry experience at AXI International doing control system design, at Insulet Corporation doing regulated documentation, and at Marmon Aerospace working with military-spec electrical assemblies. I'm based in Naples, and I'm looking to build a career in the controls and automation industry here in Southwest Florida."*

Q: What's a challenge you've overcome?

> *"I had a traumatic brain injury that knocked me out of my original career trajectory. Instead of letting it define me, I used the recovery period to teach myself programming, build an automation business, and come back stronger. That experience gave me resilience and a problem-solving mindset — when something doesn't work, I don't give up, I find another way. That same persistence is how I approach technical challenges too."*

Q: Why should we hire you over someone with more direct experience?

> *"Two things. First, I have a real foundation — three years designing schematics, building panels, and creating documentation packages. This isn't theoretical for me. Second, I learn fast and I'm hungry. I taught myself Python, AI tools, and ran my own technical consulting business. When I commit to learning something, I go deep. You'd be getting someone who has the engineering fundamentals, the documentation discipline, and the drive to get up to speed on your specific tools and workflows quickly."*

7. Questions to Ask Gil

These show genuine interest and intelligence:

1. **"What does a typical project look like for the hardware designer — from when you receive specs to when the submittal package goes out?"** (Shows you're thinking about the workflow)
2. **"What SCADA platforms and PLC brands does the team work with most?"** (Technical curiosity, helps you prep if you advance)
3. **"I saw the Cape Coral telemetry project — 274 lift stations is a massive scope. Would the hardware designer be involved in projects of that scale?"** (Shows you did research)
4. **"What does the team structure look like? Who would I be working alongside day-to-day?"** (Team fit)
5. **"What does success look like in the first 90 days for this role?"** (Goal-oriented, shows you're thinking about delivering)
6. **"Is there a particular AutoCAD Electrical workflow or template system you use for panel schematics?"** (Shows you're already thinking about the tool gap and how to close it)

8. Things to NOT Say

- Don't lead with AI/automation/Python — this is a controls hardware role, not a software role
- Don't oversell AutoCAD Electrical experience — be honest, frame the gap as closeable
- Don't mention the fitness coaching business or peptides
- Don't bash previous employers
- Don't ask about salary in a second interview unless they bring it up
- Don't say "I'm looking for remote work" — this is likely on-site/hybrid

9. Logistics Checklist

- [ ] Confirm interview time and format (in-person vs. video)
- [ ] If in-person: business casual, arrive 15 min early
- [ ] Print copies of your resume (the controls-focused version)
- [ ] Have examples ready: photos of panels you built at FGCU if available
- [ ] Download AutoCAD Electrical trial and spend 2-3 hours familiarizing
- [ ] Review UL 508A basics on YouTube (30 min)
- [ ] Review your cover letter so you remember what you claimed

- [ ] Prepare a notebook with your questions written down

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10. Salary Research

Source	Range
Control Systems Designer (FL, entry-mid)	$50,000 - $75,000
CAD Designer, industrial controls	$55,000 - $70,000
US Water parent (700+ employees)	Likely structured pay bands

Strategy: Don't bring up salary first. If asked, say: "I'm flexible and more focused on finding the right fit and growth opportunity. What's the range you have budgeted for this role?" If pressed for a number, target **$60,000-$70,000** based on your experience level and the Florida market.